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SYSTEMS AND METHODS FOR OVERRIDING AUTONOMOUS CONTROL OF A DEVICE

Abstract

Within examples, a system is described that includes a control lever for controlling operation of a device, a first actuator coupled to the control lever via a rod, and a resettable frangible link coupling a second actuator to the control lever via the rod. The resettable frangible link enables separation of coupling of the second actuator from the control lever based on an applied force to the rod by the first actuator.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/757,746, filed Jun. 28, 2024, which is a divisional of U.S. patent application Ser. No. 17/533,237 (now U.S. Pat. No. 12,049,301), filed Nov. 23, 2021, which claims priority to U.S. Provisional Application No. 63/126,774, filed on Dec. 17, 2020, the contents of which are incorporated herein by reference in their entireties.

FIELD

[0002] The present disclosure relates generally to methods to override autonomous control of a device, and more particularly, to a resettable frangible link that enables physical separation of autonomous control actuators from control levers based on an applied force.

BACKGROUND

[0003] Control systems are often commanded by either an automated actuator (in autonomous mode) or an operator (in manual mode). In some instances, the operator desires to override the automated actuator, and thus, needs an ability to regain control of inputs to the control system. In some instances, the control system includes two separate controls with one control for the human operator and another control for the autonomous control system.

[0004] In such examples, the human operator needs to have an ability to override the autonomous control, and it would be preferable to be able to return the control system to autonomous control capability once any human interaction has ended.

SUMMARY

[0005] In an example, a system is described comprising a control lever for controlling operation of a device, a first actuator coupled to the control lever via a rod, and a resettable frangible link coupling a second actuator to the control lever via the rod. The resettable frangible link enables separation of coupling of the second actuator from the control lever based on an applied force to the rod by the first actuator.

[0006] In another example, an aircraft is described comprising an engine, and a system coupled to the engine. The system comprises a control lever for controlling operation of the engine, a first actuator coupled to the control lever via a rod, and a resettable frangible link coupling a second actuator to the control lever via the rod. The resettable frangible link enables separation of coupling of the second actuator from the control lever based on an applied force to the rod by the first actuator. During autonomous operation mode of the engine, the second actuator is used to control the control lever of the engine. During manual operation mode of the engine, the first actuator is used by an operator to control the control lever of the engine. The manual operation mode is entered by the rod receiving the applied force by the first actuator causing the resettable frangible link to separate the second actuator from the rod.

[0007] In another example, a method for overriding autonomous control of a device is described. The method comprises controlling a control lever of a device via a first actuator during manual operation mode of the device. The first actuator is coupled to the control lever via a rod. The method also includes controlling the control lever of the device via a second actuator during autonomous operation mode of the device. A resettable frangible link couples the second actuator to the control lever via a rod. The method also includes entering the manual operation mode by the rod receiving an applied force via the first actuator causing the resettable frangible link to separate the second actuator from the rod.

[0008] The features, functions, and advantages that have been discussed can be achieved

independently in various examples or combined in yet other examples. Further details of the examples can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0009] The novel features believed characteristic of the illustrative examples are set forth in the appended claims. The illustrative examples, however, as well as a preferred mode of use, further objectives and descriptions thereof, will best be understood by reference to the following detailed description of an illustrative example of the present disclosure when read in conjunction with the accompanying drawings, wherein:

[0010] FIG. **1** illustrates an aircraft, according to an example implementation.

[0011] FIG. **2** illustrates a block diagram of an example of the system, according to an example implementation.

[0012] FIG. **3** illustrates a side view of an example of the system, according to an example implementation.

[0013] FIG. **4** illustrates a front view of an example of the system, according to an example implementation.

[0014] FIG. **5** illustrates a front view of another example of the system, according to an example implementation.

[0015] FIG. **6** shows a flowchart of an example of a method **200** for overriding autonomous control of the device, according to an example implementation.

[0016] FIG. **7** illustrates an example of the computing device, according to an example implementation.

DETAILED DESCRIPTION

[0017] Disclosed examples will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all of the disclosed examples are shown. Indeed, several different examples are described and should not be construed as limited to the examples set forth herein. Rather, these examples are described so that this disclosure will be thorough and complete and will fully convey the scope of the disclosure to those skilled in the art.

[0018] Within examples, a mechanically breakable control actuation device and system is described. The system allows for two separate control actuators to be used for a single controlled device, which is seen in many scenarios for redundant (primary and secondary) backup control mechanisms. For example, a throttle for an aircraft has a control rod for the pilot to use to control the throttle and another control rod that is controlled by a computer for autonomous flight. In this example, when there are two independent control mechanisms, there is a need to have an ability for one control to override the other (e.g., the pilot can override the computer control when needed). The examples herein allow the ability to disconnect (or temporally disable) one control actuator at various angles, and reconnect or reset the control actuator easily.

[0019] Referring now to the figures, FIG. **1** illustrates an aircraft **100**, according to an example implementation. The aircraft **100** includes a nose **102**, wings **104a-b**, a fuselage **106**, a tail **108**, engines **110a-b**, and a system **120** coupled to the engines **110a-b**. In examples, the system **120** is utilized for overriding autonomous control of one or more of the engines **110a-b**. Although FIG. **1** illustrates an example of a commercial passenger aircraft, other types of aircraft are used with examples described herein. In addition, depending on the type of aircraft fewer or more engines are included.

[0020] FIG. **2** illustrates a block diagram of an example of the system **120**, according to an example implementation. The system **120** includes a control lever **122** for controlling operation of a device **124**, a first actuator **126** coupled to the control lever **122** via a rod **128**, and a resettable frangible

link **130** coupling a second actuator **132** to the control lever **122** via the rod **128**. The resettable frangible link **130** enables separation of coupling of the second actuator **132** from the control lever **122** based on an applied force to the rod **128** by the first actuator **126**.

[0021] In an example, the control lever **122** controls operation of the device **124** using a bell crank movement. Thus, the control lever **122** is mechanically connected to the device **124**, and rotation of the control lever **122** controls operation of the device **124**. In examples, the device **124** includes any type of machine operated in this manner.

[0022] In an example application, the control lever **122** can include a throttle lever of the aircraft **100**, for example, and the device **124** can be an engine coupled to the throttle lever. In this example, operation of the first actuator **126** or the second actuator **132** controls operation of the engine. For instance, the first actuator **126** is a control rod for a pilot to use to control the throttle lever, and the second actuator **132** is another control rod that is controlled by a computing device **134** for autonomous operation.

[0023] In another example operation, the control lever **122** controls other aircraft flight controls, such as pitch, roll, and yaw, for example.

[0024] The first actuator **126** is permanently attached to the control lever **122** via the rod **128**. The second actuator **132** is removably attached to the control lever **122** via the resettable frangible link **130**. The first actuator **126** and the second actuator **132** each have an ability for independent control of the control lever **122**, and based on the applied force to the rod **128** by the first actuator **126**, the first actuator **126** overrides the second actuator **132**. When the first actuator **126** overrides the second actuator **132**, the second actuator **132** is disconnected from the control lever **122**. The first actuator **126** is always mechanically connected to the control lever **122**, and so an operator always has an opportunity to take over control of the device **124**.

[0025] The resettable frangible link **130** enables separation of coupling of the second actuator **132** from the control lever **122** at any rotational orientation of the control lever **122** with respect to the device **124**.

[0026] Within examples, during autonomous operation mode of the device **124**, the second actuator **132** is used to control the control lever **122** of the device **124**, and during manual operation mode of the device **124**, the first actuator **126** is used by an operator to control the control lever **122** of the device **124**. The manual operation mode is entered by the rod **128** receiving the applied force by the first actuator **126** causing the resettable frangible link **130** to separate the second actuator **132** from the rod **128**.

[0027] Generally, the second actuator **132** controls operation of the device **124** via movement of the control lever **122** according to programmed operation dictated by the computing device **134**. The second actuator **132** enables autonomous control of the device **124**. In some instances, the second actuator **132** can enter a failed state, in which cases the operator is able to regain control of the device **124** using the first actuator **126**. To do so, however, the second actuator **132** will need to be disconnected from the control lever **122** so that the second actuator **132** no longer has an ability to control the control lever **122**. The system **120** is configured such that an application of a force to the first actuator **126** causes the rod **128** to break free of the second actuator **132**. The breaking point in the system **120** is the resettable frangible link **130** that breaks and disconnects the second actuator **132** from the rod **128**, which mechanically disconnects the second actuator **132** from the control lever **122** leaving the first actuator **126** as the only actuator connected to the control lever **122** for control of the device **124**.

[0028] The breakaway characteristics can be tightly controlled to prevent erroneous fusing during normal operation and ensure complete fusing when necessary at any control input position. The system **120** is not prone to early fatigue failures, does not interfere with other systems when fused, and is resettable once a fault has been cleared.

[0029] The system **120** allows safe separation of a failed device (e.g., engine of autonomously controlled vehicle or flight controls actuator). In an example use, a single engine aircraft includes a

single operator throttle cable that attaches to a throttle lever of the engine. During autonomous mode of operation, a throttle actuator is used to control the throttle lever on the engine, backdriving the throttle cable and control lever in a cabin of the aircraft. During piloted mode, the pilot drives the throttle cable with the lever in the cabin, and the actuator is backdriven. In the event the throttle actuator jams or does not release control when requested, the pilot is able to command throttle movement through the cable, breaking free of the throttle actuator. The resettable frangible link **130** that attaches the second actuator **132** (e.g., autonomous throttle actuator) to the engine-mounted throttle lever (e.g., control lever **122**) will allow safe throttle operation in either mode, and will break free in an event where the autonomous throttle actuator has failed and the pilot must move the throttle. The resettable frangible link **130** is also be resettable when the fault has been cleared. [0030] The system **120** thus provides aircraft with two separate controls. One control is by the human pilot (e.g., the first actuator **126**) and the other control is the autonomous control (e.g., second actuator **132**). The first actuator **126** and the second actuator **132** work in tandem during autonomous control. The first actuator **126** has the ability to override the autonomous control, and is able to return to autonomous control once a situation has returned to normal.

[0031] FIG. **3** illustrates a side view of an example of the system **120**, according to an example implementation. The first actuator **126** and the second actuator **132** are operated in a linear motion to cause rotation of the control lever **122**, as shown, for a rotary input to the device **124**.

[0032] FIG. **4** illustrates a front view of an example of the system **120**, according to an example implementation. FIG. **4** illustrates further details of the resettable frangible link **130** and the rod **128**. Within examples, the resettable frangible link **130** includes a magnetically or a mechanically frangible control rod link.

[0033] In FIG. **4**, the resettable frangible link **130** includes a magnetic coupling having a first half **136** and a second half **138** to couple the second actuator **132** to the control lever **122** via the rod **128**. The first half **136** includes a base **140** for mating with a fitting **142** of the second actuator **132**. The base **140** can include a ball-type quick attach fitting and the fitting **142** can include a socket into which the ball-type quick attach fitting is inserted. The first half **136** also includes a magnet **144** positioned in the base **140**. The second half **138** includes a brace **146** attached to the rod **128**, and a magnetic pole piece **148** positioned in the brace **146**. The magnet **144** is magnetically connected to the magnetic pole piece **148**. The brace **146** includes a threaded portion for attaching to corresponding threads on the rod **128**, and the magnetic pole piece **148** is connected to the brace **146** using a screw attachment, for example.

[0034] In operation, when the first actuator **126** and the second actuator **132** both pull the control lever **122**, the resettable frangible link **130** will break under an induced bending load. For example, a magnetic force between the magnet **144** and the magnetic pole piece **148** will be overcome by the induced bending load. The second actuator **132** is then disconnected from the control lever **122** and the device **124** is operated using the first actuator **126**.

[0035] The base **140** is shown as oval to enable a breakaway at different forces (or force amounts) depending on any angle. The breakaway force may be lower at higher angles (e.g., as the fitting **142** tilts relative to the base **140**). The control lever **122** moves through an arc having a wide angle of rotation (e.g., 90 degrees or more), and thus, the resettable frangible link **130** allows for breakaway of the second actuator **132** at any orientation and at the same amount of applied force.

[0036] In other examples, the base **140** includes a footprint shape such that breakaway forces vary with output shaft angle and produce a constant force acting normal to the control lever **122**, thus, helping to ensure a constant output shaft torque regardless of shaft angular position. For example, with the base **140** in a round shape, a frangible force to break is always the same.

[0037] The resettable frangible link **130** is not required to be in line with the rod **128** (axially, or in plane). This allows for many options for mounting hardware, and can be accomplished by the base **140** being the ball and socket type configuration, where the base **140** is round and the fitting **142** mating surface is a matching round hole. Using the magnet **144** for the connection allows for easy

disconnect and quick resetting of the resettable frangible link **130**. The magnet **144** and the base **140** profile also allows for a precise control of a break point load. The shape of the base **140** can be altered from circular to adjust the break point load too.

[0038] Thus, the system **120** is resettable and is configured to break within a prescribed applied force. The breakaway force is predictable and consistent. The breakaway force is also abrupt such that there is no intermediate break position that is achieved. Thus, either the first actuator **126** or the second actuator **132** is in control of the control lever **122**. There is not a scenario where both the first actuator **126** and the second actuator **132** are in control of the control lever **122**. Once the breakaway force is achieved, the second actuator **132** is disconnected abruptly.

[0039] The magnet **144** enables the resettable frangible link **130** to break consistently with the same torque, and can be easily reset by realigning the magnet **144** with the magnetic pole piece **148**.

[0040] In some examples, the magnet **144** is a permanent magnet.

[0041] In some examples, the magnet is an electro-magnet, and the system **120** further includes a control system **150** coupled to the electro-magnet for controlling a magnetic attraction force of the electro-magnet. In FIG. 4, the control system **150** is shown coupled to a base of the first actuator **126** and wiring coupling to the magnet **144** and/or the magnetic pole piece **148** is internal to the first actuator **126**. The first actuator **126** is always connected to the rod **128**, and thus, an electrical connection between the magnet **144** and the control system **150** is always in place.

[0042] The control system **150** includes a processor, memory, and instructions stored on the memory executable by the processor to perform functions. For example, the control system **150** can be programmed to alter the magnetic attraction force of the electro-magnet based on an operating condition of the device **124**. The control system **150** can alter the magnetic force to vary a separation or break-away force that is required. In a scenario where the device **124** is an engine on an aircraft, during an approach for landing turbulence can be high, and thus, the control system **150** in this scenario alters the magnetic force to be stronger to prevent an undesired break-away caused by turbulence. Thus, a stage of flight can be used as a basis for the control system **150** to alter the magnetic force of the magnet **144** by increasing, decreasing, or otherwise changing an amount of power provided to the electro-magnet.

[0043] In another example, the control system **150** alters the magnetic force of the electro-magnet based on weather conditions in which the device **124** operates. The control system **150** can be programmed to alter the magnetic force of the electro-magnet based on any combination of factors described herein too.

[0044] In another example, the control system **150** alters the magnetic force of the electro-magnet to tailor a positional angled break away force required. In this example, a sensor **152** is included for detecting an angle of the resettable frangible link **130** with respect to the second actuator **132**, and the control system **150** alters the magnetic attraction force based on the angle. The control system **150** and electro-magnet can thus produce customized break away forces with respect to angle, as sensed by the positional sensor, which outputs signals to the control system **150** in a feedback loop to control force in position and time.

[0045] FIG. 5 illustrates a front view of another example of the system **120**, according to an example implementation. In FIG. 5, the resettable frangible link **130** a mechanically frangible control rod **154** to couple the second actuator **132** to the control lever **122** via the rod **128**. The first half **136** still includes the base **140** for mating with the fitting **142** of the second actuator **132**, and the second half **138** includes the brace **146** attached to the rod **128**. The mechanically frangible control rod **154** connects the first half **136** to the second half **138**.

[0046] The mechanically frangible control rod **154** has a first end **156** and a second end **158**, and each of the first end **156** and the second end **158** is threaded such that the first end **156** attaches to the base **140** and the second end **158** attaches to the brace **146**.

[0047] In operation, when the first actuator **126** applies a force to the rod **128**, the mechanically

frangible control rod **154** will break, based on the force being above a threshold amount, and the second actuator **132** will then be disconnected from the control lever **122**.

[0048] In one example, the mechanically frangible control rod **154** is a tensile element. The mechanically frangible control rod **154** has a predefined preload and will break under a certain applied pressure. To reset the resettable frangible link **130**, the broken fragments of the mechanically frangible control rod are unscrewed from both sides (e.g., from the base **140** and from the brace **146**), and a new tensile element is inserted.

[0049] FIG. **6** shows a flowchart of an example of a method **200** for overriding autonomous control of the device **124**, according to an example implementation. Method **200** shown in FIG. **6** presents an example of a method that could be used with the aircraft **100** shown in FIG. **1**, with the system **120** shown in FIGS. **2-5**, or with the computing device **134** shown in FIG. **7**, for example. Further, devices or systems are used or configured to perform logical functions presented in FIG. **6**. In some instances, components of the devices and/or systems are configured to perform the functions such that the components are actually configured and structured (with hardware and/or software) to enable such performance. In other examples, components of the devices and/or systems are arranged to be adapted to, capable of, or suited for performing the functions, such as when operated in a specific manner. Method **200** includes one or more operations, functions, or actions as illustrated by one or more of blocks **202-206**. Although the blocks are illustrated in a sequential order, these blocks may also be performed in parallel, and/or in a different order than those described herein. Also, the various blocks may be combined into fewer blocks, divided into additional blocks, and/or removed based upon the desired implementation.

[0050] It should be understood that for this and other processes and methods disclosed herein, flowcharts show functionality and operation of one possible implementation of present examples. In this regard, each block or portions of each block may represent a module, a segment, or a portion of program code, which includes one or more instructions executable by a processor for implementing specific logical functions or steps in the process. The program code is be stored on any type of computer readable medium or data storage, for example, such as a storage device including a disk or hard drive. Further, the program code can be encoded on a computer-readable storage media in a machine-readable format, or on other non-transitory media or articles of manufacture. The computer readable medium may include non-transitory computer readable medium or memory, for example, such as computer-readable media that stores data for short periods of time like register memory, processor cache and Random Access Memory (RAM). The computer readable medium may also include non-transitory media, such as secondary or persistent long term storage, like read only memory (ROM), optical or magnetic disks, compact-disc read only memory (CD-ROM), for example. The computer readable media may also be any other volatile or non-volatile storage systems. The computer readable medium may be considered a tangible computer readable storage medium, for example.

[0051] In addition, each block or portions of each block in FIG. **6**, and within other processes and methods disclosed herein, may represent circuitry that is wired to perform the specific logical functions in the process. Alternative implementations are included within the scope of the examples of the present disclosure in which functions may be executed out of order from that shown or discussed, including substantially concurrent or in reverse order, depending on the functionality involved, as would be understood by those reasonably skilled in the art.

[0052] At block **202**, the method **200** includes controlling the control lever **122** of the device **124** via the first actuator **126** during manual operation mode of the device **124**, and the first actuator **126** is coupled to the control lever **122** via the rod **128**.

[0053] At block **204**, the method **200** includes controlling the control lever **122** of the device **124** via the second actuator **132** during autonomous operation mode of the device **124**, and the resettable frangible link **130** couples the second actuator **132** to the control lever **122** via the rod **128**.

[0054] At block **206**, the method **200** includes entering the manual operation mode by the rod **128** receiving an applied force via the first actuator **126** causing the resettable frangible link **130** to separate the second actuator **132** from the rod **128**.

[0055] Within an example, the resettable frangible link **130** includes a magnetic coupling having the first half **136** and the second half **138** to couple the second actuator **132** to the control lever **122** via the rod **128**, and the method **200** further includes realigning the first half **136** and the second half **138** of the magnetic coupling causing a magnetic attachment of the first half **136** and the second half **138** in order to reattach the second actuator **132** to the rod **128** for reentering the autonomous operation mode of the device **124**. In an example, the realignment is performed by a manual attachment. In another example, the realignment occurs by magnetic reattachment of the magnet **144** to the magnetic pole piece **148**.

[0056] In FIGS. 2-5, the system **120** is shown to include a computing device **134**. In examples, the control system **150** takes the form of a computing device too. FIG. 7 illustrates an example of the computing device **134**, according to an example implementation.

[0057] To perform the functions described, the computing device **134** includes one or more processors **160**, instructions **162** stored on non-transitory computer readable media **164**, a communication interface **166**, an output interface **168**, and each component of the computing device **134** is connected to a communication bus **170**. The computing device **134** also includes hardware to enable communication within the computing device **134** and between the computing device **134** and other devices (not shown). The hardware may include transmitters, receivers, and antennas, for example.

[0058] In examples, the communication interface **166** is a wireless interface and/or one or more wireline interfaces that allow for both short-range communication and long-range communication to one or more networks or to one or more remote devices. Such wireless interfaces provide for communication under one or more wireless communication protocols, Bluetooth, WiFi (e.g., an institute of electrical and electronic engineers (IEEE) 802.11 protocol), Long-Term Evolution (LTE), cellular communications, near-field communication (NFC), and/or other wireless communication protocols. Such wireline interfaces include an Ethernet interface, a Universal Serial Bus (USB) interface, or similar interface to communicate via a wire, a twisted pair of wires, a coaxial cable, an optical link, a fiber-optic link, or other physical connection to a wireline network. Thus, in examples, the communication interface **166** is configured to receive input data from one or more devices, and is also configured to send output data to other devices.

[0059] The non-transitory computer readable media **164** includes or take the form of memory, such as one or more computer-readable storage media that can be read or accessed by the one or more processors **160**. The computer-readable storage media can include volatile and/or non-volatile storage components, such as optical, magnetic, organic or other memory or disc storage, which can be integrated in whole or in part with the one or more processors **160**. The non-transitory computer readable media **164** is considered non-transitory computer readable media. In some examples, the non-transitory computer readable media **164** can be implemented using a single physical device (e.g., one optical, magnetic, organic or other memory or disc storage unit), while in other examples, the non-transitory computer readable media **164** can be implemented using two or more physical devices.

[0060] The non-transitory computer readable media **164** thus is a computer readable medium, and the instructions **162** are stored thereon. The instructions **162** include computer executable code.

[0061] In examples, the one or more processors **160** are general-purpose processors or special purpose processors (e.g., digital signal processors, application specific integrated circuits, etc.). The one or more processors **160** receive inputs from the communication interface **166** as well as outputs from other sensors (e.g., the sensor **152**), and process them to generate outputs that are stored in the non-transitory computer readable media **164**. The one or more processors **160** can be configured to execute the instructions **162** (e.g., computer-readable program instructions) that are stored in the

non-transitory computer readable media **164** and are executable to provide the functionality of the computing device **134** described herein.

[0062] The output interface **168** outputs information for reporting or storage, and thus, the output interface **168** is similar to the communication interface **166** and can be a wireless interface (e.g., transmitter) or a wired interface as well.

[0063] Note that although this disclosure has described use of the methods and systems for use on aircraft, the same functions apply equally to use of the methods and system on board any type of vehicle in order to change from an autonomous to a manual mode of operation. The methods and systems can also find use within non-vehicles or stationary areas to change operation of any type of machine.

[0064] By the term “substantially” and “about” used herein, it is meant that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.

[0065] Different examples of the system(s), device(s), and method(s) disclosed herein include a variety of components, features, and functionalities. It should be understood that the various examples of the system(s), device(s), and method(s) disclosed herein include any of the components, features, and functionalities of any of the other examples of the system(s), device(s), and method(s) disclosed herein in any combination or any sub-combination, and all of such possibilities are intended to be within the scope of the disclosure.

[0066] The description of the different advantageous arrangements has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the examples in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different advantageous examples describe different advantages as compared to other advantageous examples. The example or examples selected are chosen and described in order to best explain the principles of the examples, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various examples with various modifications as are suited to the particular use contemplated.

Claims

1. A system comprising: a control lever for controlling operation of a device; a first actuator coupled to the control lever via a rod, wherein the first actuator is configured to control the control lever in a first operation mode of the device; and a second actuator coupled to the control lever via the rod and a resettable frangible link, wherein the second actuator is configured to control the control lever in a second operation mode of the device.
2. The system of claim 1, wherein the resettable frangible link includes a magnetic coupling having a first half and a second half to couple the second actuator to the control lever via the rod.
3. The system of claim 2, wherein the magnetic coupling is a permanent magnet.
4. The system of claim 1, wherein the resettable frangible link enables separation of coupling of the second actuator from the control lever based on an applied force to the rod by the first actuator.
5. The system of claim 1, wherein: the control lever is a throttle lever, the device is an engine coupled to the throttle lever, and operation of the first actuator or the second actuator controls operation of the engine.
6. The system of claim 1, wherein the control lever controls operation of the device using a bell crank movement.
7. The system of claim 1, wherein: the first operation mode is a manual operation mode, and the second operation mode is an autonomous operation mode.
8. The system of claim 1, wherein the resettable frangible link enables separation of coupling of the

second actuator from the control lever at any rotational orientation of the control lever with respect to the device.

9. The system of claim 1, wherein: the control lever is a throttle lever for an aircraft, the first actuator is a first control rod for a pilot to use to control the throttle lever, and the second actuator is a second control rod that is controlled by a computing device for autonomous operation.

10. The system of claim 1, wherein the first actuator is permanently attached to the control lever via the rod.

11. The system of claim 1, wherein: the first actuator and the second actuator each have an ability for independent control of the control lever, and the first actuator is configured to override the second actuator based on an applied force to the rod by the first actuator.

12. The system of claim 1, wherein the resettable frangible link comprises a mechanically frangible control rod to couple the second actuator to the control lever via the rod.

13. The system of claim 1, wherein the resettable frangible link comprises: a base for mating with a fitting of the second actuator; a brace attached to the rod; and a mechanically frangible control rod having a first end and a second end, wherein each of the first end and the second end is threaded such that the first end attaches to the base and the second end attaches to the brace.

14. The system of claim 1, wherein: the resettable frangible link includes a magnetic coupling having a first half and a second half to couple the second actuator to the control lever via the rod, and the first half and the second half of the magnetic coupling are configurable to be realigned causing a magnetic attachment of the first half and the second half to reattach the second actuator to the rod for reentering the second operation mode of the device.

15. The system of claim 1, wherein the resettable frangible link includes an electro-magnet, the system further comprising: a control system coupled to the electro-magnet for controlling a magnetic attraction force of the electro-magnet.

16. The system of claim 15, wherein the control system is configured to alter the magnetic attraction force based on an operating condition of the device.

17. The system of claim 15, further comprising: a sensor for detecting an angle of the resettable frangible link with respect to the second actuator, wherein the control system is configured to alter the magnetic attraction force based on the angle.

18. An aircraft, comprising: an engine; and a system coupled to the engine, the system comprising: a control lever for controlling operation of the engine; a first actuator coupled to the control lever via a rod, wherein the first actuator is configured to control the control lever in a first operation mode of the engine; and a second actuator coupled to the control lever via the rod and a resettable frangible link, wherein the second actuator is configured to control the control lever in a second operation mode of the engine.

19. The aircraft of claim 18, wherein: the first operation mode is a manual operation mode, and the second operation mode is an autonomous operation mode.

20. The aircraft of claim 18, wherein: the control lever is a throttle lever for the aircraft, the first actuator is a first control rod to enable a pilot to control the throttle lever, and the second actuator is a second control rod that is controlled by a computing device for autonomous operation.
